Autonomous Decision-Making Method for UUV Emergency Management Based on Graph Retrieval-Augmented Generation Large Language Model

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***Abstract*—When operating with minimal or no human supervision, underwater unmanned vehicles often encounter unforeseen emergencies originating from the external environment, internal subsystems, or task interactions. These abrupt and uncertain events pose serious threats to vehicle safety. Conventional emergency-response methods that depend on static rule bases are limited to predefined, deterministic situations and are therefore inadequate for complex, time-varying scenarios. To overcome this limitation, we propose an autonomous emergency-management framework that couples a domain knowledge graph with a retrieval-augmented generative large language model (LLM). First, we construct an Unmanned Underwater Vehicle (UUV) emergency-management knowledge graph encompassing environmental factors, internal components, and multi-level cascading faults. Second, we develop a clustering-based retrieval mechanism that provides the model with traceable, fine-grained decision context. Third, through the joint use of prompt engineering and LoRA fine-tuning, we obtain a lightweight decision model deployable on edge devices. Finally, scenario-driven temporal simulations validate the framework's reasoning accuracy and control effectiveness. Comparative results indicate that our approach delivers broader information coverage, higher decision accuracy, and faster response times than traditional rule-based baselines, thereby markedly enhancing UUV safety and reliability in challenging marine environments.**

Keywords—Unmanned underwater vehicles; knowledge graph; Large model decision-making; autonomous emergency control

I. INTRODUCTION (*HEADING 1*)

Unmanned Underwater Vehicles (UUVs) are autonomous, intelligent platforms designed for underwater operations. During missions, they are vulnerable to environmental factors—such as ocean currents, stratification layers, and submerged obstacles—as well as to sudden malfunctions of onboard sensors or actuators, any of which may drive the vehicle away from its nominal operating state [1]. In these situations, human-in-the-loop emergency control often cannot respond quickly or effectively enough. Onboard autonomous emergency management is therefore essential for true UUV autonomy and for improving overall mission performance [2]. However, conventional emergency-decision schemes that rely on static rule bases lack the adaptability required to cope with complex, evolving environments and cascading internal faults. A promising solution is to augment UUV planning with semantic inference [3], as explored in the studies that follow, with a focus on ease of use.

Patron P, Miguelanez, et al. at the Marine Systems Laboratory, Heriot-Watt University (UK), proposed a hierarchical semantic knowledge model in which semantic information is exchanged among UUV subsystems, enabling the vehicles to respond more effectively to unexpected events [4][5].

Li Hangyu et al. [6] of Harbin Engineering University proposed constructing a semantic knowledge network that captures the logical relationships among different UUV subsystems, thereby enabling knowledge sharing and reuse while also supporting reasoning-based decision making.

Yao Hongfei, Benyin Li, et al.[7][8] of Harbin Engineering University built a Bayesian network derived from a semantic knowledge network and, on this basis, developed an influence-diagram–based utility model to support emergency decision making in UUV operations.

K. Agyei et al. [9] at the University of Hertfordshire (UK) developed a risk-aware control framework that leverages large language models (LLMs) to enhance collision-avoidance and trajectory-tracking performance in autonomous surface vehicles.

Muhayy Ud Din et al. [10] proposed an innovative task planning framework that leverages LLMs to enhance the autonomous decision-making capabilities of unmanned surface vehicles (USVs) in dynamic marine environments.

A. Sezgin et al. [11] at the Siemens Istanbul R&D Center proposed a centralized decision-making framework that couples LLMs with retrieval-augmented generation (RAG) to improve real-time adaptability and mission reliability within the Internet of Drones (IoD).

A KG is a structured semantic network that represents concepts, entities, and their relationships as subject–predicate–object triples, thereby offering high extensibility and machine-readable semantics [12][13]. In fault diagnosis and decision support, KG-based methods have become a key research focus.

Conversely, because LLMs have shown strong capabilities for semantic generalization and reasoning in unseen scenarios, researchers are increasingly integrating LLMs with

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knowledge graphs to mitigate hallucinations and close the knowledge gaps that arise during LLM-based reasoning [14].

Jiashuo Sun et al. [15] of Xiamen University tackled the poor performance of LLMs on complex knowledge-reasoning tasks by proposing "Think on Graph," an algorithmic framework that leverages external knowledge graphs.

Previous studies lay a solid foundation for applying LLMs to decision-making in unmanned systems. With retrieval-augmented generation now mature, LLMs can be coupled with external data sources to process complex inputs, adapt to changing environments, and perform semantic reasoning. In UUV emergency-management scenarios, this capability promises greater autonomy and smarter decisions.

This paper focuses on constructing a UUV fault-model library, designing compact decision-making LLMs, and establishing a collaboration mechanism between knowledge graphs and LLMs. We develop a graph-search-enhanced generative framework for UUV emergency reasoning and introduce a lightweight LLM—refined through prompt engineering and distillation—to satisfy the platform's power and compute constraints. The proposed approach aims to deliver more accurate and reliable UUV emergency strategies.

II. Classification and Diagnosis of UUV Emergency Incidents

A. Classification of UUV Emergency Incidents Using Situational Information

In a knowledge-graph-based decision-making system, the initial task is to organize domain knowledge into a coherent, machine-readable structure. The UUV emergency management knowledge graph transforms uncertain situational data into clear semantic representations through emergency-event ontology construction, enabling emergency event identification. [16]. UUV situational data fall into three main categories: mission information, external marine-environment information, and internal self-state information [17]. The internal state can be further decomposed into two dimensions—navigation state and platform state. Data for each dimension are collected by the corresponding onboard sensors, as illustrated in Figure 1.

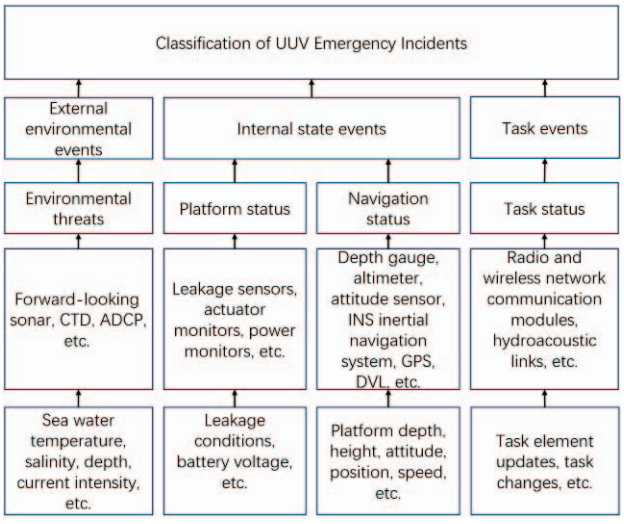


Fig. 1. Classification of UUV emergency safety incidents

B. Identification of Emergency Incidents in UUVs

Based on the situational data of the unmanned-underwater vehicle, we identify multiple categories of emergency events. To support the development of corresponding control strategies, selected quantitative parameters are translated into qualitative levels that indicate different severities of failure. Representative emergency events are summarized in Table 1.

TABLE I. Emergency Events Based on Situational Information.

Environmental	Platform	Navigation	Mission
Light loss	Abnormal battery drain	Dangerous depth	Mission payload anomaly
Lack of optical features	Low battery	Dangerous height	Mission payload failure
Dynamic underwater target	Extremely low power	Attitude anomaly.	Communication anomaly
Static underwater target	Propeller anomaly.	Position anomaly.	Communication failure.

Environmental events are chiefly detected using exteroceptive sensor data. Navigation-related events are identified by comparing key navigation state parameters—such as heading, depth, altitude, and position—against predefined safety thresholds. Mission-related events are detected from payload health data and communication-link metrics.

Platform-level events are diagnosed by the onboard health-monitoring system. Some events can be inferred directly from status data—for example, the remaining-battery percentage is used to flag "abnormal power decline," "low power," or "critical power." Incipient faults in key actuators such as thrusters and control surfaces, however, are not readily observable in raw status data. To detect these early-stage failures, this paper proposes a multiplicative-fault observer that applies gain-thresholds to achieve real-time actuator monitoring and emergency-event detection. The design of the observer is outlined below:

$$\begin{cases} \dot{\hat{\omega}} = -J^{-1}\omega^{\times} J\omega + J^{-1}\Gamma\hat{p} + J^{-1}\Omega \\ \dot{\hat{p}} = \lambda\Gamma^{\mathrm{T}} J^{-\mathrm{T}}\tilde{\omega} - \varepsilon\hat{p} \end{cases} \tag{1}$$

In this equation, $\hat{\omega} \in R^{3\times1}$ and $\hat{p} \in R^{3\times1}$ denote the estimated angular velocity and the estimated multiplicative actuator fault, respectively. λ and ε are positive constants. $\tilde{\omega} = \omega - \hat{\omega}$ represents the observer-estimation error, and Ω is the sliding-mode term introduced for disturbance attenuation, defined as follows:

$$\Omega = \alpha\tilde{\omega} + v\,\mathrm{sign}(\tilde{\omega}) \tag{2}$$

In this equation, α is a positive constant. $\mathrm{sign}(\tilde{\omega})$ and v are defined as follows:

$$\mathrm{sign}(\tilde{\omega}) = \left[\mathrm{sign}\left(\tilde{\omega}_1\right) \quad \mathrm{sign}\left(\tilde{\omega}_2\right) \quad \mathrm{sign}\left(\tilde{\omega}_3\right)\right]^{\mathrm{T}} \tag{3}$$

$$\dot{v} = \begin{cases} \beta\,|\,\tilde{\omega}\,| - \gamma\left(v - v^{*}\right) & v \le v^{*} \\ 0 & v > v^{*} \end{cases} \tag{4}$$

In this equation, β and γ are positive constants. Define $d \in R^3$ as the three-axis disturbance torque vector, with $\bar{d}$ as its upper bound. Define $v^{*} \ge \bar{d}$ to guarantee effective disturbance suppression. The dynamic estimation error of the angular velocity is given by:

$$\dot{\tilde{\omega}} = J^{-1}\Gamma\tilde{p} + J^{-1}d - J^{-1}\Omega \tag{5}$$

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C. Probabilistic Analysis for Event Determination

Fault tree analysis (FTA) is a probabilistic system–fault modelling technique that quantifies the influence of basic events on higher-level system emergencies [18]. Drawing on earlier UUV FTA studies, we estimate the probability that situationally identified emergency events will escalate into more severe events or lead to UUV capability loss. If the probability that a single basic event—or a combination of basic events—triggers a higher-level emergency exceeds 0.7, the corresponding causal relation is inserted into the knowledge graph as a logical edge. Representative emergency events incorporated in this way are listed in Table 2.

TABLE II. DERIVED EMERGENCY EVENT.

Platform events	Navigational events	Task events
Structural damage	Collision avoidance failure	Mission failure
UUV lost	Relative navigation failure.	
	Absolute navigation failure.	

D. Determining the Emergency Response Level

Fault states are classified into four urgency levels. Exploiting the predictive-reasoning capability of the knowledge graph, we devise emergency control measures for two temporal scenarios: events that may be triggered by the current emergency (time point t_1) and events that have already materialized (time point t_2). The urgency levels are defined as follows:

- Level 1. The vehicle must offload its tasks at both t_1 and t_2.
- Level 2. Targeted mitigation is required at t_1, followed by either task offloading or a return to port at t_2.
- Level 3. Targeted mitigation is carried out at t_2; if the fault cannot be cleared within a specified time, the vehicle must return to port.
- Level 4. The event can be ignored or handled with minor adjustments at t_2, as circumstances permit.

Using task-related emergencies as an example, Table 3 presents the level definitions and their corresponding control measures.

TABLE III. CLASSIFICATION OF EMERGENCY MANAGEMENT LEVELS FOR TASK EVENTS

Level	Event	time point t_1	time point t_2
1	Mission failure	Jettison or return to base	Jettison or return to base
2	Mission payload failure.	Enable redundancy	Jettison or return to base
3	Mission payload anomaly	——	Enable redundancy
4	Communication anomaly	——	Decelerate

III. DEVELOPING AN EMERGENCY DECISION SYSTEM POWERED BY KNOWLEDGE GRAPHS

The emergency decision-making process for UUVs is divided into three stages: situational awareness, threat assessment, and collaborative decision-making. The framework developed in this study is depicted in Figure 2.

The UUV emergency management system leverages situational awareness data to detect and characterize emerging incidents. Once an emergency—whether familiar or novel—is identified, the decision-making module queries the knowledge graph for decision-support elements such as event-propagation relationships and corresponding response protocols, and then synthesizes an appropriate strategy. After execution, the system monitors key performance indicators to assess the plan's effectiveness. Newly validated emergency-management knowledge is subsequently incorporated into the knowledge graph, continuously refining future decisions. The remainder of this section describes the graph's construction and update mechanism.

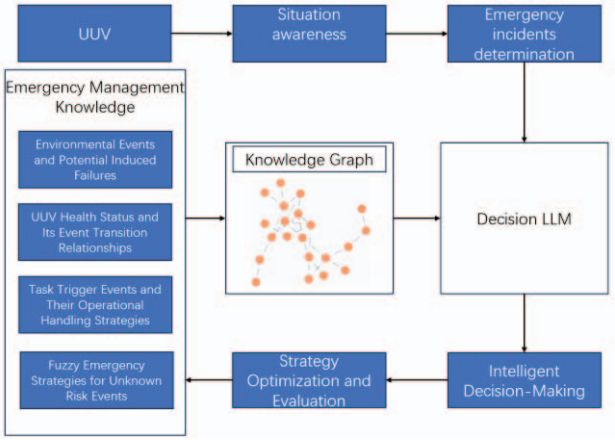


Fig. 2. UUV Emergency Management Decision-Making System

A. Knowledge graph construction

Building on the established UUV fault and emergency-response frameworks, we construct a knowledge graph through a bottom-up methodology. Emergency events are modelled as entities, fault-propagation routes as relations, and corresponding control measures as attributes. These elements form subject–predicate–object triples, and the resulting knowledge is stored on a per-event basis. Figure 3 presents the storage schema for emergency-event entities.

```
{
  "name":"【Event Name】",
  "description": "【Emergency Management Strategies】",
  "type": "EVENT"
}
```

Fig. 3. Graph Data Storage Template

Once the base-level knowledge graph has been constructed, we introduce "clusters"—a structural concept from complex-network science—to enhance the specificity and interpretability of knowledge retrieval for the LLM decision engine [19]. Treating the entire graph as an undirected, weighted network, we employ the Leiden algorithm [20] to partition entities while preserving semantic coherence. The algorithm maximizes modularity, producing clusters whose internal connection density is markedly higher than their external connectivity, thereby exposing genuine association patterns.

In a UUV-emergency-management context, each cluster represents a set of tightly coupled, mutually influential fault states [21]. By analyzing the transition paths within a cluster, we generate an "intra-cluster association report" for decision support. During query time, this report is injected as prior knowledge, enabling the LLM to pinpoint events most likely to escalate the present fault scenario and anticipate additional threats that could trigger future escalations. The result is a more informed and proactive decision-making process. Figure

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4 illustrates the storage schema of the intra-cluster association report.

```
{
 "title": "【Cluster Structure Name】",
 "summary": "【Cluster Structure Overview】",
 "findings": [
  {
   "summary": "【Cluster Intra-Status Report of Emergency Events】",
   "explanation": "【Causes and Explanations】"
  },
 ]
}
```

Fig. 4. Cluster Report Case

Upon receiving a UUV fault report, the retrieval module identifies relevant anchors in the knowledge graph—such as associated entities, relations, attributes, and cluster-analysis reports. It then ranks and filters these candidate knowledge items according to their relevance, ultimately supplying the decision-making model with a curated context for reference.

B. Graph update

To support continual evolution and self-learning of the knowledge graph, we adopt a performance-driven update policy. If a response measure for a known fault fails on several consecutive occasions, the system revises the corresponding entity's attributes and removes that ineffective measure. When an UUV encounters an unfamiliar fault state, the system sequentially executes candidate response plans. If a particular plan successfully eliminates the fault, a new triple is inserted into the graph and classified under Level 3 emergency controls. This policy endows the knowledge graph with dynamic-update capability, ensuring ongoing optimization of the decision engine.

IV. Design of an LLM-Based Intelligent Decision-Making System for UUV Emergency Safety

UUV platforms represent edge devices with severe limits on on-board computation and power, leaving little headroom for large decision-making models. Deploying a full-scale LLM on such hardware is therefore impractical [22]. To satisfy these constraints, the framework employs DeepSeek-R1—a distilled, lightweight inference model derived from Llama-3.1-8B as its decision-making engine.

A. Prompt Template Design

Prompt engineering encompasses techniques for crafting and refining input prompts to improve the quality of LLM outputs and better align them with user requirements [23]. The prompt templates employed by the decision-making module are summarized in Table 4.

TABLE IV. Prompt Template

Prompt Content	Prompt Goal
Character	Limit the way LLM understands input
Goal	Prompt the content of the output
Output Template	Limit the format of LLM output
Rule	Prohibit fabrication
Data Table	Provide data for presenting the knowledge graph

B. Fine-Tuning of the Decision-Making Model

Fine-tuning a LLM consists of retraining a pre-trained network on task- or domain-specific data so that it performs more accurately and efficiently within that scope. In this work, we apply Low-Rank Adaptation (LoRA) to fine-tune the distilled base model for the UUV emergency-decision task [24]. The training dialogues are crafted to reflect emergency-control scenarios and embed an explicit chain-of-thought rationale that guides the model's reasoning, and follow a dedicated answer template that guarantees complete, standardized outputs and mitigates hallucinations.

The base model is fine-tuned for four epochs over a corpus of 2 000 dialogue samples, using a micro-batch size of 8 and gradient-accumulation steps of 4 (effective batch size = 32). A linear learning-rate scheduler starts at 1×10^{-4} with 50 warm-up steps. Training is performed in FP16 mixed precision to speed up computation and reduce GPU memory usage. To meet real-time inference demands, the resulting model is quantized to INT8, which boosts throughput while preserving decision accuracy for emergency-control tasks.

C. Integrated Decision-Making with Large Language Models and Knowledge Graphs

Figure 5 illustrates the integration mechanism between the decision-making LLM and the knowledge graph

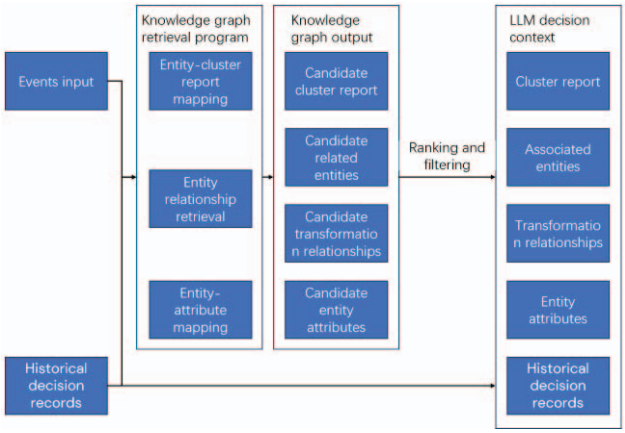


Fig. 5. The interaction between LLM and knowledge graphs

The UUV Emergency-Control Knowledge Graph supplies contextual information to the LLM decision engine. Each incoming emergency event is first converted into an embedding (a semantic vector) by an embedding model. The retrieval module then queries the vector index to identify entities, relations, attributes, and clustered incident reports whose embeddings lie close to that of the input event in the high-dimensional semantic space. Finally, the cosine similarity between the input-event embedding and each candidate embedding is computed as follows:

$$\cos(\theta) = \frac{\mathrm{A} \cdot \mathrm{B}}{\| \mathrm{A} \| \ \| \mathrm{B} \|} \tag{6}$$

Vector A denotes the semantic embedding retrieved from the knowledge base, whereas vector B denotes the embedding of the input event. Their Euclidean norms, $\| \mathrm{A} \|$ and $\| \mathrm{B} \|$, respectively describe the magnitudes of A and B When the cosine-similarity score approaches 1, the retrieved knowledge is considered highly relevant to the query. Candidate items are subsequently ranked in descending order of similarity, and the top results are supplied to the LLM as contextual input for decision making.

The decision-making LLM employs COT reasoning during the process. First, it evaluates the impact on the current UUV state based on the attributes of the current event entity and related cluster reports. Next, it infers possible migration

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routes based on related entities and their relationships. Finally, it categorizes decisions into three levels, aiming to suppress the most critical fault events, maintain system stability, and restore normal operating conditions or terminate the task.

V. Verification of Simulation Results

Referring to the simulation experiment setup in the study on autonomous decision-making methods for UUVs by Hongfei et al. [25], we assume that our UUV is currently performing a regional search task at a speed of 2 m/s. The specific task involves surveying the seabed terrain within 3–5 nautical miles of the mother ship platform, collecting various types of sensor data, and transmitting this data back to the mother ship platform for backup. After analyzing and processing the information detected over a certain period, we selected 10 time points of situational information, with the emergency event determination results demonstrated in Figure 6. The decision results of the proposed system at time point 3 are illustrated in Figure 7.

As can be seen from the system output, the decision-making model successfully made decisions based on the fine-tuned template. After comparing the decision results with the knowledge graph, it was confirmed that the decision-making LLM effectively utilized the information provided by the knowledge graph for reasoning. Assuming that all emergency events persist until time 4.

When the new emergency event "rudder malfunction" occurs, the system searches for the fault transfer chain related to this event. Given that the fault states of "Static Water Target" and "Mission Payload Abnormal" are still present, when the "Rudder Abnormal" fault threatens the UUV's ability to avoid obstacles, the system prioritizes the control of rudder fault tolerance as the primary measure based on the rule of "prioritizing related faults," while abandoning monitoring of the current mission payload and enabling redundancy. Later, at 7:00, due to the reoccurrence of "task load anomaly," the system determines that "task load failure" could be directly triggered by the current fault state and executes the final decision to terminate the mission, initiate a return to base, or jettison the payload, following the previously established contingency plan.

In this scenario, the average cosine similarity values between the decision support information within the decision-making LLM context and the corresponding input emergency event semantic vector are shown in Table 5.

TABLE V. SIMILARITY STATISTICS

Events input	Cosine similarity values
Task payload anomaly	0.90
underwater target	0.85
Strong ocean current	0.86
Servo anomaly	0.83

These scores indicate that the decision support information provided by the knowledge graph is highly relevant to the current emergency situation, thereby effectively expanding the depth of knowledge of the decision-making LLM through the use of the knowledge graph and enhancing its reasoning capabilities.

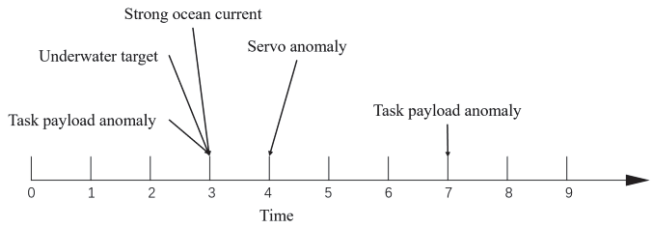


Fig. 6. Time point sequence

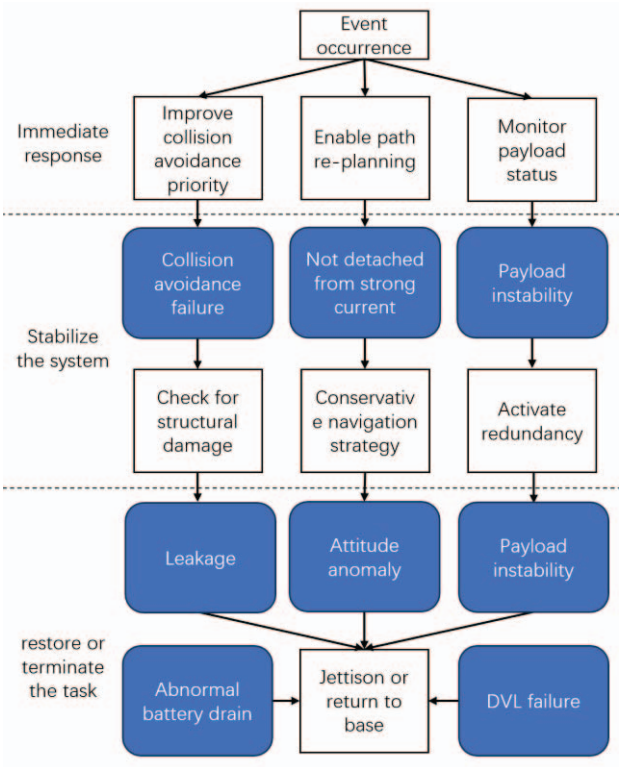


Fig. 7. Output at time point 3

Two traditional decision-making methods, Random Forests and Influence Diagrams, were introduced and compared against the framework proposed in this paper. Multiple experiments were conducted with different input emergency events at time 3 to determine whether the emergency response decisions for various events align with expert recommendations for the corresponding scenarios. The accuracy rate of the control response for each emergency event was calculated using the following formula.

$$\text{Accuracy} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{7}$$

In the formula, where TP represents an emergency response that is consistent with expert recommendations, FP represents an incorrect response, and FN indicates that no response was generated. Some of the statistical results are summarized in Table 6.

TABLE VI. DECISION ACCURACY STATISTICS

Event	sample size	LLM	RF	IF
Task payload anomaly.	200	95.5%	82.0%	84.5%
underwater target	150	94.0%	80.0%	84.0%
Strong ocean current	140	92.9%	78.6%	81.4%
Servo anomaly	140	94.2%	72.4%	83.2%

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thrust vector anomaly	150	94.7%	73.2%	84.2%
low battery	160	96.2%	81.3%	81.9%

The results indicate that the framework proposed in this paper demonstrates superior performance compared to traditional classifiers in the classification and coordinated handling of multiple emergency events. By doing so, it can more comprehensively integrate decision-making knowledge and generate more targeted strategies for complex and uncertain emergency scenarios.

VI. Conclusions and Future Work

Addressing the challenges of emergency management decision-making for UUVs in complex marine environments, this paper proposes a UUV emergency management reasoning and decision-making framework based on graph search-enhanced generative LLMs. It constructs a UUV emergency management knowledge graph and a lightweight decision-making LLM adapted to the UUV environment, and designs knowledge retrieval and presentation methods to enable their integration. Through cosine similarity evaluation and comparative experiments with traditional decision-making methods, the effectiveness of the proposed method is validated systematically. However, while the introduction of the knowledge graph effectively addresses the limitations of LLM decision-making frameworks in terms of specialized knowledge acquisition and information consistency, there is still room for improvement in template adaptability and strategy-to-instruction conversion mechanisms. Therefore, future research will focus on optimizing model pre-training strategies and deep integration between LLMs and agents, aiming to further enhance the precision and generalization capabilities of decision-making intelligence.

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